

## Review Article

## Psychoeducation for relapse prevention in bipolar disorder: a systematic review of efficacy in randomized controlled trials

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**Objectives:** Previous reviews have concluded that interventions including psychoeducation are effective in preventing relapse in bipolar disorder, but the efficacy of psychoeducation itself has not been systematically reviewed. Our aim was to evaluate the efficacy of psychoeducation for bipolar disorder in preventing relapse and other outcomes, and to identify factors that relate to clinical outcomes.

**Methods:** We employed the systematic review of randomized controlled trials of psychoeducation in participants with bipolar disorder not in an acute illness episode, compared with treatment-as-usual, and placebo or active interventions. Pooled odds ratios (ORs) for non-relapse into any episode, mania/hypomania, and depression were calculated using an intent-to-treat (ITT) analysis, assigning dropouts to relapse, with a sensitivity analysis in which dropouts were assigned to non-relapse (optimistic ITT).

**Results:** Sixteen studies were included, eight of which provided data on relapse. Although heterogeneity in the data warrants caution, psychoeducation appeared to be effective in preventing any relapse [ $n = 7$ ; OR: 1.98–2.75; number needed to treat (NNT): 5–7, depending on the method of analysis] and manic/hypomanic relapse ( $n = 8$ ; OR: 1.68–2.52; NNT: 6–8), but not depressive relapse. Group, but not individually, delivered interventions were effective against both poles of relapse; the duration of follow-up and hours of therapy explained some of the heterogeneity. Psychoeducation improved medication adherence and short-term knowledge about medication. No consistent effects on mood symptoms, quality of life, or functioning were found.

**Conclusions:** Group psychoeducation appears to be effective in preventing relapse in bipolar disorder, with less evidence for individually delivered interventions. Better understanding of mediating mechanisms is needed to optimize efficacy and personalize treatment.

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Although pharmacotherapy is recognized as the mainstay of therapy for bipolar disorder, outcomes for many patients are suboptimal (1). Combining medication with psychological approaches may be a cost-effective strategy (2) and there has been much interest in recent years in psychoeducation as an adjunctive treatment for preventing relapse.

The term *psychoeducation* has been defined as ‘any intervention that educates patients and their families about their illness with a view to improving their long-term outcome’ (3). This can range from simply providing information on medication to enhance adherence (4), to intensive, complex interventions covering drug and illness information,

stressors, coping strategies, lifestyle management, and personalized relapse plans (5). The target of the intervention can vary from a focus on educating the person with bipolar disorder, to a family focus, or even only involving carers. There is considerable overlap between psychoeducation and specific psychotherapies such as cognitive behavioral therapy (CBT), interpersonal and social rhythm therapy (IPSRT), and family-focused therapy (FFT) (see *Supplementary Fig. 1* and further discussion in *Methods*). It is therefore not surprising that studies included in recent reviews of psychoeducation for bipolar disorder (3, 6, 7) have had considerable overlap with those in broader reviews of psychological therapies for relapse prevention (8, 9).

In spite of the above difficulties, there are core elements of psychoeducation which are often included in, but are not the defining aspects of, other specific psychological therapies. These are education about the illness, the importance of regular routines and medication adherence, early detection of warning signs of relapse with strategies to prevent progression into full episodes, and enhancement of general coping strategies (10). Reviews of psychoeducation for bipolar disorder, while concluding that it is effective in preventing relapse (3, 6, 7), have commented that more evidence is needed that psychoeducation itself is effective, rather than as part of a multicomponent approach (6), and that the *active ingredient* and level of expertise needed for effective delivery are unclear (3).

In the present review, we assess the efficacy of randomized, controlled trials of psychoeducation for patients with bipolar disorder who are not in an acute episode, in preventing relapse and other outcomes, and attempt to identify which components relate to efficacy.

## Methods

For the present review, we defined psychoeducation as: (i) a discrete psychological intervention involving primarily the patient with bipolar disorder; (ii) providing information about bipolar disorder and/or its treatment; and (iii) relating this information to aiding self-management of the disorder.

**Inclusion criteria:** randomized controlled trials (RCTs) of psychoeducation against treatment-as-usual (TAU) or a control intervention for patients with bipolar disorder not in an acute illness episode. Interventions that were designed to control for non-specific effects of treatment (such as non-directive group meetings) were deemed a placebo

control and analyzed together with TAU; control interventions presumed to be effective treatments were viewed as active controls.

**Exclusion criteria:** studies using therapies with additional modality-specific features that distinguished them from psychoeducation, as identified by Miklowitz et al. (11); these were CBT (focusing on cognitive restructuring and self-rated mood charts), IPSRT (focusing on regulation of sleep/wake cycles and daily routine), and family treatments (focusing on family communication and skills training) (see *Supplementary Fig. 1*). For the same reason, we excluded collaborative care studies in which psychoeducation was a part of a multifaceted intervention involving changes to service delivery interventions. We also excluded studies in which psychoeducation was used as an acute treatment or included patients in an acute phase of illness, if the primary diagnosis was not bipolar disorder, or if the target was patients with bipolar disorder with a comorbid diagnosis such as personality disorder or substance misuse.

The search strategy used the keywords *bipolar disorder* or *manic depress\** or *mania* and *education* or *psychoeducation* or *relapse prevention* in Medline, Embase, Psycinfo, and Cochrane databases, with no starting date up to 19 September 2013, limiting papers to clinical trials or reviews or meta-analyses, English language, and human subjects (see *Supplementary material* for full search criteria). Papers and review articles were hand-searched to identify further studies.

*Supplementary Figure 2* presents a flowchart showing the search results. Briefly 1,522 unique papers were identified, of which 107 described psychological interventions (trial design or outcome) in bipolar disorder and 57 concerned psychoeducation studies as defined above. Intervention outcomes were reported from 37 trials (48 papers), of which 16 trials (24 papers) were RCTs of psychoeducation administered outside an illness episode and 21 trials (24 papers) were retrospective, open, non-comparative, non-randomized studies or RCTs of psychoeducation given during an acute illness episode or irrespective of phase of illness. The last group was not included in the present review (listed in *Supplementary Table 1*).

Data were extracted using standardized proformas by both authors. The interventions were examined to identify the content and process, delivery (therapist details, group versus individual, internet versus face-to face), and the number and length of intervention sessions. Details of the interventions are summarized in *Supplementary Table 2* and study details and outcomes in Table 1. The risk of bias in the studies was assessed using the

Cochrane Collaboration risk of bias tool ([www.cochrane-handbook.org](http://www.cochrane-handbook.org)) (see *Supplementary Figs. 3 and 4*).

The efficacy of psychoeducation in preventing relapse was analyzed quantitatively by calculating the pooled odds ratio (OR) of not relapsing into a mood episode in the psychoeducation compared with the placebo control/TAU; there were insufficient similar studies to pool data on psychoeducation compared with active treatment. Further information on numbers relapsing was requested, where necessary, from authors. Given the potential risk from attrition bias due to differential dropouts in treatment and control arms in many studies (see *Supplementary Figs. 3 and 4*), the primary analysis was a conservative full intent-to-treat (ITT) analysis in which, for studies not reporting ITT outcomes, all dropouts were assumed to have relapsed. The robustness of the results was tested by two sensitivity analyses. First, given that it is extremely unlikely that every participant dropping out relapsed, especially not into both poles of illness, an *optimistic* ITT was carried out in which it was assumed that dropouts remained well; the real result therefore lies between these conservative and optimistic ITT analyses. Second, a *restricted* analysis was carried out of the conservative ITT dataset which excluded the internet study (18) and the studies by Colom et al. (5, 20) – the former because the lack of a therapist contact means therapy delivery is uncertain, the latter because the specialist setting and very high relapse rates (> 90%) in the control condition raise questions about the generalizability of the results. The key outcome was the number of patients not relapsing from the start of intervention to the end of follow-up into: (i) any episode; (ii) an episode of elevated mood (hypo-manic, manic, or mixed episode), hereafter termed a manic episode, for brevity; and (iii) a depressive episode. For one study (13), data were available only for the polarity of the first relapse, and for two studies (5, 20) mixed-episode relapses were reported separately and could not be included. The acceptability of treatment was assessed by comparing dropouts from each arm during treatment where the psychoeducation and control treatment conditions were of comparable length, using full ITT of all participants randomized. One study (20) reported no dropouts and one was added to each arm for the OR calculation.

A variance-weighted random-effects pooled OR was calculated using the meta-analysis function in StatsDirect ([www.statsdirect.com/](http://www.statsdirect.com/)). Publication bias was examined using the Horbold–Egger test (28). Statistical heterogeneity was examined using

the  $I^2$  statistic (29) and explored if above 25% by looking for outliers and study characteristics. Method of delivery (group versus individual), duration of follow-up, and hours of therapy were highly inter-related (see below) so that the primary subgroup analysis was limited to the method of delivery. If substantial heterogeneity remained, exploratory analyses by duration of follow-up (up to one year versus > 1 year) and hours of therapy (below versus above median split) were carried out. To aid clinical interpretation, we present the percentage of patients relapsing in each arm as an unweighted average across studies not corrected for study duration, and for significant results the pooled number needed to treat (NNT) or needed to harm (NNH) calculated from the pooled risk difference (random effects) and rounded to the nearest whole number.

It was not possible to pool other outcomes that are described narratively, and the influence of intervention content and study design were explored where possible.

## Results

The details of the 16 RCTs are given in Table 1. The authors of six studies provided further information, five on relapse numbers (13, 15, 18, 19, 22), one on patient characteristics (22), and a further author confirmed separate patient populations in the two studies by Colom et al. (5, 20). There was an uncertain risk of selection and attrition bias in a number of studies but only a minority had a high risk of bias overall (see *Supplementary Figs. 3 and 4*, and *Comments column* Table 1). Studies varied in the degree and duration of remission at entry, and the size of most of the studies was small (median 70 participants). In most studies, patients met DSM-IV criteria for bipolar disorder. Two studies recruited only patients with bipolar I disorder and 11 studies reported the percentage of patients with bipolar II disorder (ranging from 2% to 34%). Comparison conditions were TAU (seven studies), a placebo control condition (six studies, three matched for therapist time), or an active treatment (four studies). One study had both a TAU and an active comparison arm (26). Psychoeducation was delivered in group format in eight studies, individually in five studies, combined group and individual format in two studies, and by internet in one study.

### Nature of the psychoeducation interventions

*Supplementary Table 2* presents the details of the individual interventions. There were broadly four

Table 1. Summary of randomized controlled studies of psychoeducation for participants with bipolar disorder not in an acute episode

| Study <sup>a</sup>                                    | Population                                                                 | Psychoeducation (PE)                                                       | Comparator (C) | Number/dropouts from study assessment <sup>b</sup> | Duration from start of intervention (weeks) | Relapse (PE versus C)                                 | Other outcomes (PE compared with C unless stated otherwise)                                                           | Quality/ comments                                                                                                                                                                         |
|-------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------|----------------------------------------------------|---------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dogan and Sabanciogullari 2003 (12)                   | DSM-IV BP on long-term lithium therapy                                     | Individual and group education about BP and lithium 3 sessions (3–4 hours) | TAU            | PE: 16/2<br>C: 14/4                                | 12                                          | –                                                     | PE: improved mood symptoms, QoL, lithium levels, and medication knowledge and regularity in taking it<br>C: no change | <ul style="list-style-type: none"> <li>• Allocation by alternate numbers;</li> <li>• Blinding not stated;</li> <li>• Completer analysis of each group separately</li> </ul>               |
| D'Souza et al. 2010 (13)                              | BP in remission, discharged from hospital in last month; 14% BP-II         | Group to companion –patient dyad 12 x 90-min sessions (18 hours)           | TAU            | PE: 27/7<br>C: 31/6                                | 60                                          | Increased time to relapse<br>Fewer patients relapsing | Improved mania ratings and medication adherence<br>No difference in depressive symptoms                               | <ul style="list-style-type: none"> <li>• LOCF analysis;</li> <li>• Relapse: hospitalization/intensive community support (first episode only)</li> </ul>                                   |
| Javadpour et al. 2013 (14)                            | BP with recurrent episodes, in full remission following hospital discharge | Individual 8 x 50-min sessions + monthly telephone contact (10 hours)      | TAU            | PE: 45/9<br>C: 41/13                               | 86                                          | Fewer relapses and hospitalizations                   | Less symptomatic deterioration, improved QoL and medication adherence                                                 | <ul style="list-style-type: none"> <li>• Analysis population unclear;</li> <li>• Relapse: no longer meeting remission criteria</li> </ul>                                                 |
| Lobban et al. 2010 (15)                               | Clinical diagnosis BP with recurrent episodes, remission $\geq$ 1 month    | Individual 6 x 60 min (6 hours)                                            | TAU            | PE: 56/6<br>C: 40/0                                | 48                                          | No difference in time to relapse or number relapsing  | Improved social and occupational functioning                                                                          | <ul style="list-style-type: none"> <li>• Cluster randomized by community team;</li> <li>• Blinding failed;</li> <li>• Completer analysis;</li> <li>• Relapse: new mood episode</li> </ul> |
| Peet and Harvey 1991 (4)<br>Harvey and Peet 1991 (16) | Lithium clinic attenders with affective disorder                           | Individual and group 3 sessions (3 hours)                                  | TAU            | PE: 30/2<br>C: 30/0                                | 8                                           | –                                                     | Increased knowledge, improved attitude to lithium, fewer missed doses<br>No difference in lithium levels              | <ul style="list-style-type: none"> <li>• Randomization method not stated;</li> <li>• Not blinded;</li> <li>• Self-rated measures of knowledge and attitudes</li> </ul>                    |

Table 1. (Continued)

| Study <sup>a</sup>      | Population                                                       | Psychoeducation (PE)                          | Comparator (C)                       | Number/dropouts from study assessment <sup>b</sup> | Duration from start of intervention (weeks) | Relapse (PE versus C)                                                                         | Other outcomes (PE compared with C unless stated otherwise)                                                                                                                                                                                       | Quality/ comments                                                                                                                                                                                                               |
|-------------------------|------------------------------------------------------------------|-----------------------------------------------|--------------------------------------|----------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Perry et al. 1999 (17)  | DSM-III-R BP, ≥ 2 relapses (one in last year); 9% BP-II          | Individual 7–12 x 60 min (7–12 hours)         | TAU                                  | PE: 34/1<br>C: 35/0                                | 78                                          | Mania: increased time to relapse and fewer relapses<br>Depression: no difference in relapse   | <ul style="list-style-type: none"> <li>Improved social and occupational functioning;</li> <li>Increased antidepressant use;</li> <li>No difference in episode duration, prescriptions, drug levels, hospitalization, community contact</li> </ul> | <ul style="list-style-type: none"> <li>Randomization by minimization on key features;</li> <li>Blinding not stated;</li> <li>ITT analysis;</li> <li>Relapse: ≥ 5 days syndromal criteria</li> </ul>                             |
| Smith et al. 2011 (18)  | DSM-IV BP remitted ≥ 3 months; 12% BP-II                         | Individual by internet 8 modules (8–12 hours) | TAU                                  | PE: 24/7<br>C: 26/6                                | 42                                          | No difference in manic and depressive episodes                                                | No difference in QoL, functioning, insight, mood symptoms                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Dynamic block randomization balanced for key features;</li> <li>Complete analysis;</li> <li>Self-rated, not blind;</li> <li>Relapse: syndromal episode</li> </ul>                        |
| <b>Placebo control</b>  |                                                                  |                                               |                                      |                                                    |                                             |                                                                                               |                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                 |
| Castle et al. 2010 (19) | DSM-IV BP not in acute episode; 25% BP-II                        | Group 12 x 90 min + 3 boosters (23 hours)     | Weekly telephone calls               | PE: 42/10<br>C: 42/2                               | 52                                          | Fewer overall and manic relapses<br>No difference in depressive relapses                      | Less time unwell in any and depressive relapse<br>No difference in mood symptoms scores                                                                                                                                                           | <ul style="list-style-type: none"> <li>Randomized stratified for number of episodes;</li> <li>Analysis by intervention completers;</li> <li>Not blind for relapse;</li> <li>Relapse: &gt; defined rating scale value</li> </ul> |
| Colom et al. 2003 (20)  | DSM-IV BP-I, euthymic ≥ 6 months, good compliance for ≥ 6 months | Group 21 x 90 min (32 hours)                  | Matched non-directive group sessions | PE: 25/0<br>C: 25/0                                | 104                                         | Increased time to any recurrence; fewer manic and depressive recurrences and hospitalizations | –                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Randomized stratified for number of episodes;</li> <li>ITT/LOCF analysis;</li> <li>Relapse: &gt; defined rating scale value</li> </ul>                                                   |

6 Table 1. (Continued)

| Study <sup>a</sup>                               | Population                                                          | Psychoeducation (PE)                         | Comparator (C)                          | Number/dropouts from study assessment <sup>b</sup>                           | Duration from start of intervention (weeks) | Relapse (PE versus C)                                                               | Other outcomes (PE compared with C unless stated otherwise)                                                  | Quality/ comments                                                                                                                                                                           |
|--------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Colom et al. 2003 (5)<br>Colom et al. 2009 (21)  | DSM-IV BP-I and BP-II, euthymic<br>≥ 6 months;<br>17% BP-II         | Group<br>21 × 90 min<br>(32 hours)           | Matched non-directive group sessions    | PE: 60/4 at 2 years,<br>6 at 5 years<br>C: 60/0 at 2 years,<br>11 at 5 years | 104<br>260                                  | Fewer recurrences of all types and hospitalizations; increased time to recurrences  | Higher lithium levels at 2 years                                                                             | <ul style="list-style-type: none"> <li>Randomized stratified for number of episodes;</li> <li>ITT analysis;</li> <li>Relapse: &gt; defined rating scale value</li> </ul>                    |
| De Barros Pellegrinelli et al. 2013 (22)         | DSM-IV BP, euthymic<br>≥ 1 month;<br>22% BP-II                      | Group<br>16 × 90 min twice weekly (24 hours) | Matched relaxation group sessions       | PE: 32/13<br>C: 23/14                                                        | 60                                          | No differences in relapse or hospitalizations                                       | Improved patient-rated CGI, no difference symptom ratings, function, QoL                                     | <ul style="list-style-type: none"> <li>Randomized 'matched' by key features;</li> <li>ITT/LOCF analysis;</li> <li>Relapse: &gt; remission cutoff on rating scales</li> </ul>                |
| Eker and Harkin 2012 (23)                        | DSM-IV BP, in remission, 'able to learn'                            | Group<br>6 × 90–120 min (9–12 hours)         | Individual brief medication explanation | PE: 36/6<br>C: 35/3                                                          | 12                                          | –                                                                                   | Improved adherence in PE group, no change in C<br>Trend to improved medication and adherence attitudes in PE | <ul style="list-style-type: none"> <li>Randomized 'according to' key features;</li> <li>Completer/LOCF analysis;</li> <li>Not blinded</li> </ul>                                            |
| <b>Active control</b><br>Parikh et al. 2012 (24) | Recurrent DSM-IV BP, remitted ≥ 1 month; 31% BP-II                  | Group<br>6 × 90 min (9 hours)                | Individual CBT                          | PE: 109/48<br>C: 95/32                                                       | 78                                          | No difference in time to recurrence                                                 | No difference in symptom ratings, social adjustment                                                          | <ul style="list-style-type: none"> <li>Randomized in permuted blocks;</li> <li>ITT survival analysis/growth curve modeling;</li> <li>Recurrence: &gt; defined rating scale value</li> </ul> |
| Rea et al. 2003 (25)                             | DSM-III-R BP-I, after discharge following hospitalization for mania | Individual<br>21 × 30 min (11 hours)         | FFT                                     | PE: 25/5<br>C: 28/6                                                          | 104                                         | No difference in risk of relapse, increased hospitalizations, or number of relapses | No difference in medication adherence                                                                        | <ul style="list-style-type: none"> <li>Randomization method not stated;</li> <li>ITT survival analysis;</li> <li>Relapse: &gt; defined rating scale value</li> </ul>                        |



Table 1. (Continued)

| Study <sup>a</sup>        | Population                                 | Psychoeducation (PE)                                      | Comparator (C)              | Number/dropouts from study assessment <sup>b</sup> | Duration from start of intervention (weeks) | Relapse (PE versus C)    | Other outcomes (PE compared with C unless stated otherwise)                                                                                                                                      | Quality/ comments                                                                                                                                                                    |
|---------------------------|--------------------------------------------|-----------------------------------------------------------|-----------------------------|----------------------------------------------------|---------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Torrent et al. 2013 (26)  | DSM-IV BP, remission ≥ 3 months; 17% BP-II | Group 21 × 90 min (32 hours)                              | (i) TAU<br>(ii) FR          | PE: 82/20<br>C1: 80/14<br>C2: 77/22                | 26                                          | –                        | <ul style="list-style-type: none"> <li>• FR improved functioning compared with TAU;</li> <li>• PE not different to other arms;</li> <li>• No differences in neurocognitive measures</li> </ul>   | <ul style="list-style-type: none"> <li>• Computer-generated randomization;</li> <li>• ITT (LOCF) analysis</li> </ul>                                                                 |
| Zaretsky et al. 2008 (27) | DSM-IV BP not in an episode; 34% BP-II     | Individual 7 sessions (duration not defined) (7–10 hours) | Individual CBT following PE | PE: 39/19<br>C: 40/14                              | 52                                          | No difference in relapse | <ul style="list-style-type: none"> <li>• More depressed days/month</li> <li>• No differences in dysfunctional attitudes, social functioning, QoL, mood symptoms, medication adherence</li> </ul> | <ul style="list-style-type: none"> <li>• Randomization stratified by illness episodes;</li> <li>• Completer analysis;</li> <li>• Relapse: &gt; defined rating scale value</li> </ul> |

BP = bipolar disorder; BP-I = bipolar I disorder; BP-II = bipolar II disorder; C = comparator; CBT = cognitive behavioral therapy; CGI = clinical global improvement; FFT = family-focused therapy; FR = functional remediation; IPSRT = interpersonal and social rhythm therapy; ITT = intent-to-treat; LIFE = life interval follow-up evaluation; LOCF = last observation carried forward; PE = psychoeducation; QoL = quality of life; TAU = treatment-as-usual.

<sup>a</sup>Grouped by nature of comparator.

<sup>b</sup>Dropouts from intervention completion are analyzed separately (see text).

overlapping areas of information provided: (i) causes, symptoms, and course of bipolar disorder; (ii) types and details of treatments, including drugs and psychological therapies; (iii) relapse patterns, including prodromes and triggers such as early symptoms, drugs and alcohol, life and interpersonal stresses, and circadian and social rhythms; and (iv) strategies to minimize the risk of relapse, including scheduling activity, regularizing sleep and social rhythms, avoiding or managing triggers, treatment adherence, and gender-specific aspects. Added to information there was a process/personalization aspect consisting of relating the information to personal histories; developing personalized coping strategies including lifestyle changes, early warning signs, and action plans; and how to obtain help.

Other potential therapeutic elements were the process of group delivery, offering the opportunity for sharing experience (30, 31), and the inclusion of caregivers in one study (13).

Some earlier studies concentrated on only limited aspects of psychoeducation such as lithium knowledge and adherence (4, 12, 16) or early warning signs (17), whereas later studies mostly used a more comprehensive approach, with one-half of the studies explicitly based upon the interventions of Colom and Vieta (32) or Bauer and McBride (33). The number of sessions varied from two to 21, ranging from internet delivered to group sessions with experienced psychologists. The degree of personalization varied from a few didactic sessions (4, 12, 16) to being highly interactive (5, 20, 22).

#### The effect of psychoeducation on relapse

*Psychoeducation against placebo control/TAU.* Nine RCTs reported data on patients relapsing (Table 1); it was not possible to extract data for quantitative analysis from one study (14), an individually delivered intervention against TAU with an 18-month follow-up, which reported significantly fewer average hospitalizations and relapses in the intervention arm. Seven studies reported the number of patients relapsing into any episode, and eight reported depressive and manic relapse separately. In the studies that could be pooled, type of delivery (group versus individual), comparison (placebo control versus TAU), the duration of study follow-up (median 60 weeks, range: 42–104 weeks), and hours of therapy (median 20.5 hours, range: 6–32 hours) were inter-related; the five group studies included all four studies with placebo control and had the longest follow-up and greatest number of hours of therapy. Length of

study follow-up and hours of therapy tended to correlate positively ( $r = 0.70$ ,  $p = 0.051$ ).

*Any relapse:* In the control condition, 30% of participants did not relapse compared with 45% of those who received psychoeducation (unweighted mean across all studies); respective figures for the optimistic ITT were 40% and 60%. In the pooled analysis, this difference significantly favoured psychoeducation (Fig. 1), with moderate heterogeneity between studies but no evidence of selection bias [OR = 1.98, 95% confidence interval (CI): 1.09–3.58,  $p = 0.024$ ,  $I^2 = 54\%$ , Horbold–Egger  $p = 0.91$ ]. Sensitivity analyses did not substantially alter the results: (i) optimistic ITT: OR = 2.75, 95% CI: 1.42–5.33,  $p = 0.003$ ,  $I^2 = 53\%$ ; (ii) restricted analysis:  $n = 4$ , OR = 1.64, 95% CI: 1.02–2.63,  $p = 0.04$ ,  $I^2 = 0\%$ .

Studies with group delivery were effective and relatively homogeneous ( $n = 5$ , OR = 2.80, 95% CI: 1.63–4.82,  $p < 0.001$ ,  $I^2 = 19\%$ ), whereas, individually, delivery was not effective ( $n = 2$ , OR = 0.89, 95% CI: 0.45–1.76,  $p = 0.74$ ) (Fig. 1). NNTs to prevent any relapse were seven (95% CI: 4–25) for all studies and four (95% CI: 3–7) for group studies; the equivalent figures for the optimistic ITT were five (95% CI: 3–15) and four (95% CI: 3–11). The NNT for the restricted analysis was eight (95% CI: 4–111).

*Manic relapse:* In the control condition, 54% of participants did not have a manic or hypomanic relapse compared with 69% of those who had received psychoeducation (unweighted mean);

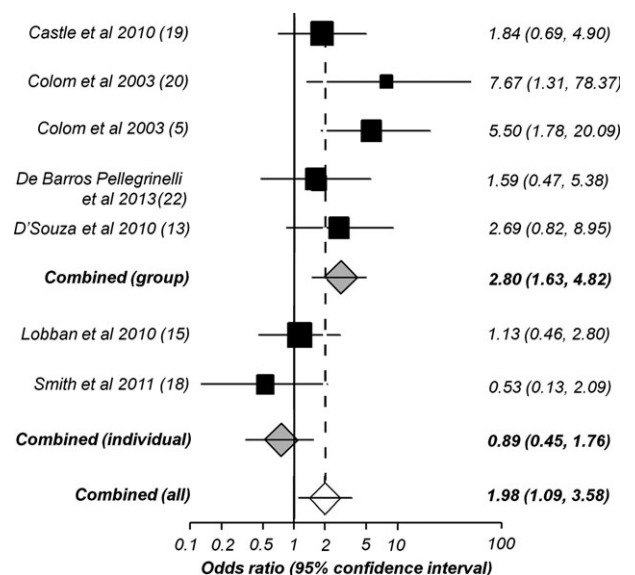


Fig. 1. Pooled analysis of any relapse comparing psychoeducation with placebo control or treatment-as-usual. Odds ratio of not relapsing into any episode. Values > 1 favor psychoeducation.



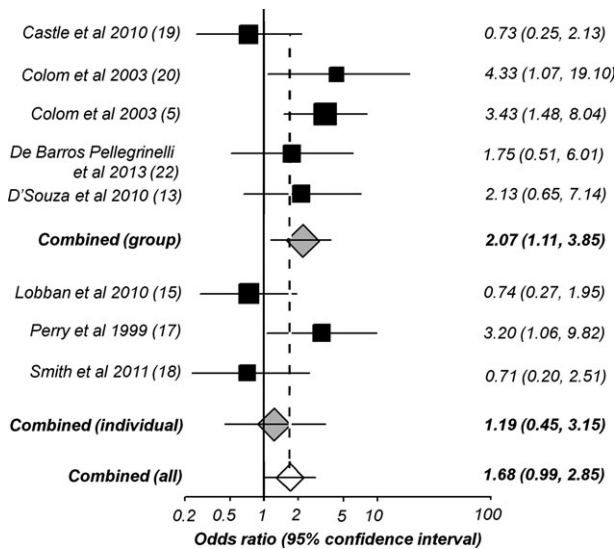


Fig. 2. Pooled analysis of manic/hypomanic relapse comparing psychoeducation with placebo control or treatment-as-usual. Odds ratio of not relapsing into a manic/hypomanic episode. Values > 1 favor psychoeducation.

respective figures for the optimistic ITT were 64% and 84%. This difference in favor of psychoeducation just missed significance (Fig. 2), with moderate heterogeneity between studies but no evidence of selection bias (OR = 1.68, 95% CI: 0.99–2.85,  $p = 0.06$ ,  $I^2 = 55\%$ , Horbold–Egger  $p = 0.67$ ). Sensitivity analyses showed a significant effect for the optimistic ITT: OR = 2.52, 95% CI: 1.69–3.76,  $p < 0.001$ , with low heterogeneity ( $I^2 = 0\%$ ), but for not the restricted analysis:  $n = 5$ , OR = 1.41, 95% CI: 0.77–2.55,  $p = 0.26$ ,  $I^2 = 44\%$ .

Studies with group delivery were effective but moderate heterogeneity remained ( $n = 5$ , OR = 2.07, 95% CI: 1.11–3.85,  $p = 0.02$ ,  $I^2 = 47\%$ ); individually delivered studies were not effective, with moderate-to-high heterogeneity ( $n = 3$ , OR = 1.19, 95% CI: 0.45–3.15,  $p = 0.72$ ,  $I^2 = 65\%$ ) (Fig. 2). Given the persisting heterogeneity, this was further explored and was best explained by follow-up duration; psychoeducation was effective in studies with a duration >1 year, with low heterogeneity ( $n = 4$ , OR = 3.16, 95% CI: 1.04–5.15,  $p < 0.001$ ,  $I^2 = 0\%$ ) but not in those with follow-up to one year ( $n = 4$ , OR = 0.88, 95% CI: 0.53–1.45,  $p = 0.61$ ,  $I^2 = 0\%$ ).

NNTs to prevent manic relapse were eight (95% CI: 4– $\infty$ ) for all studies, six (95% CI: 3–39) for group studies, and four (95% CI: 3–7) for longer duration studies; equivalent figure for the optimistic ITT were six (95% CI: 4–19), six (95% CI: 3–40), and four (95% CI: 3–7).

**Depressive relapse:** In the control condition, 57% of participants did not have a depressive

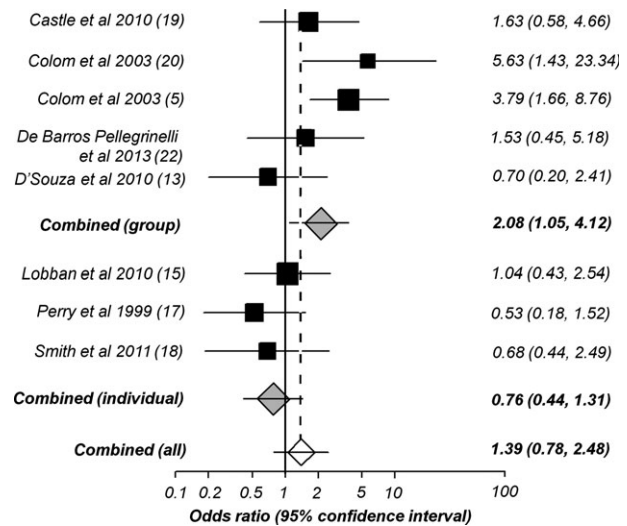


Fig. 3. Pooled analysis of depressive relapse comparing psychoeducation with placebo control or treatment-as-usual. Odds ratio of not relapsing into a depressive episode. Values > 1 favor psychoeducation.

relapse compared with 66% of those who had received psychoeducation (unweighted mean); respective figures for the optimistic ITT were 66% and 81%. This difference was not significant (Fig. 3), with moderate-to-high heterogeneity between studies and no evidence of selection bias (OR = 1.39, 95% CI: 0.78–2.48,  $p = 0.26$ ,  $I^2 = 63\%$ , Horbold–Egger  $p = 0.39$ ). Sensitivity analyses showed a borderline significant effect for the optimistic ITT, with heterogeneity remaining moderate to high: OR = 1.95, 95% CI: 1.00–3.80,  $p = 0.051$ ,  $I^2 = 61\%$ , but no effect in the restricted analysis:  $n = 5$ , OR = 1.00, 95% CI: 0.65–1.53,  $p = 0.99$ ,  $I^2 = 0\%$ .

Studies with group delivery were effective but moderate-to-high heterogeneity remained ( $n = 5$ , OR = 2.08, 95% CI: 1.05–4.12,  $p = 0.04$ ,  $I^2 = 57\%$ ); individually delivered studies were not effective ( $n = 3$ , OR = 0.76, 95% CI: 0.44–1.31,  $p = 0.32$ ,  $I^2 = 0\%$ ) (Fig. 3). Further exploration of heterogeneity showed that this was best explained by the number of hours of therapy; psychoeducation was effective in studies with more than 20 of hours therapy, with low-to-moderate heterogeneity ( $n = 4$ , OR = 2.68, 95% CI: 1.49–4.81,  $p = 0.001$ ,  $I^2 = 30\%$ ), but it was not effective in those with fewer than 20 hours of therapy ( $n = 4$ , OR = 0.75, 95% CI: 0.46–1.22,  $p = 0.24$ ,  $I^2 = 0\%$ ).

NNTs to prevent depressive relapse were 12 (95% CI: 5 to NNH 18, i.e. not statistically significant) for all studies, six (95% CI: 3–77) for group studies, and five (95% CI: 3–11) for studies with more than 20 hours of therapy; equivalent figures

for the optimistic ITT were nine (95% CI: 5–∞), six (95% CI: 3–111), and five (95% CI: 3–∞).

*Psychoeducation against active control.* Two studies compared group (24) or individual (27) psychoeducation against individual CBT and reported no significant difference in relapse rates. A third study, comparing individual psychoeducation with FFT, found no difference in the number who relapsed but relapses were fewer and of lesser severity in the latter (25) (Table 1).

*Acceptability of psychoeducation.* There were data available on participant discontinuation during the treatment period from eight RCTs: four against placebo control (5, 19, 20, 22), three against active treatment (24, 25, 27), and one against an active treatment and TAU (26). More participants receiving psychoeducation dropped out (28%) than those in the control condition (23%) (unweighted mean), but this was a non-significant difference, with low heterogeneity between studies and no significant selection bias (OR = 1.25, 95% CI: 0.84–1.85,  $p = 0.27$ ,  $I^2 = 22\%$ , Horbold-Egger  $p = 0.72$ ) (Supplementary Fig. 5).

Effect of psychoeducation on severity of mood symptoms

*Psychoeducation against placebo control/TAU.* Of the six studies reporting outcomes (Table 1), three found no difference between treatment arms over time in observer/self-reported manic or depressive symptoms (18, 19, 22). Three studies against TAU reported a benefit from psychoeducation in mood symptoms (12–14), but the pattern was inconsistent and not statistically robust in one (12).

*Psychoeducation against active control.* In two studies comparing psychoeducation and CBT, there was no difference in observer-rated manic symptoms between groups (Table 1); one found a non-significant trend to lower depression scores after CBT (27), while the other found no difference (24).

Effect of psychoeducation on functioning and quality of life

Seven studies reported outcomes against placebo control/TAU (Table 1). Four studies used the same self-report quality-of-life measure, with the outcome varying from a benefit for psychoeducation (14), benefit only in different subdomains of the measure (12, 18), or no difference (22). Of five studies assessing functioning, two reported an

improvement with psychoeducation (15, 17) and three did not (18, 22, 26).

In two studies against CBT (Table 1), the treatment groups did not differ in functioning, adjustment, or satisfaction with life (24, 27) and no difference in functioning was found compared with functional remediation (26).

Effect of psychoeducation on medication knowledge, attitudes, and adherence

Three studies against TAU or a brief information condition assessed knowledge and/or attitudes to medication (Table 1). In two studies, knowledge about lithium was increased at 3–4 months after baseline in the intervention group (4, 12) and in two studies attitudes to medication improved in the psychoeducation group at 3–6 months after baseline (4, 23).

Adherence was measured by various methods in seven studies against placebo/TAU (Table 1). Of four studies reporting drug levels, lithium levels were increased/improved in two group studies (5, 12) and unchanged in the other two using individual delivery (16, 17); no difference in levels of other mood-stabilizing drugs was reported in two studies (5, 17). Five studies assessed the taking of medication, with all reporting greater adherence (12–14, 16, 23). A further two-year study measured adherence but did not report this outcome (5).

In two active comparator studies, no difference in adherence was found comparing psychoeducation with CBT or FFT (25, 27).

Other outcomes

Psychoeducation did not alter neurocognitive functioning on a neuropsychological test battery compared with TAU or functional remediation in one study (26) and was inferior to CBT in altering dysfunctional negative beliefs in another (27).

## Discussion

The main finding was that psychoeducation appears to be effective in preventing any, and manic/hypomanic, relapse in bipolar disorder, with less certainty for depressive relapse. The findings were robust with regard to assumptions about dropouts, given that the effect must lie between those found in the conservative and optimistic ITT analyses; however, caution is required because of the limited number of studies analyzed and their heterogeneity. Limiting analysis to group psychoeducation reduced study heterogeneity for any relapse, increased the sizes of effect, and showed

efficacy also in preventing depressive relapse. The sparse data comparing psychoeducation with specific psychotherapies suggest equal efficacy in preventing any relapse occurring but possibly a poorer illness or symptom course compared with CBT or FFT. As measured by intervention completion, psychoeducation appears to have comparable acceptability to control interventions, including specific psychotherapies.

Data are patchy with regard to other outcomes but there is reasonable evidence that psychoeducation increased medication adherence, although inconsistency about whether it improves mood symptoms or quality of life in patients already in remission.

## The nature of psychoeducation related to outcome

Although group delivery appears to be more effective than individual psychoeducation, the interrelationship between group delivery, longer study duration, and a greater number of hours of therapy means that it is not possible to be certain that group delivery is the key factor. This is particularly the case for depressive relapse (see below). In addition, we were unable to include in the quantitative analysis a study using individual psychoeducation which reported positive results (14). The potential benefit of group delivery is that other members of the group are an additional source of information, support, and sharing of experience, which may provide vicarious learning and reduce feelings of stigma (30, 31). No effect of group delivery is apparent in other outcomes, apart from improving serum lithium levels, which was seen only in group studies (5, 12).

There is reasonable evidence that psychoeducation improves reported medication adherence, although this is less consistent when objective blood measures are used. There appears to be a short-term improvement in medication knowledge and attitudes, further supported by a non-comparative study which found that increased knowledge about lithium persisted to two years (34). The relationship between medication adherence and prevention of relapse is not clear. A two-year study found higher, more stable serum lithium levels after psychoeducation (35), but the effect was small and not found with other drugs (5); the same research group found psychoeducation to be effective in highly medication-adherent patients (20). This suggests that, while improved medication adherence might contribute to relapse prevention after psychoeducation, it does not appear to be the main factor.

Although it is not possible to disentangle the number of hours of therapy from group delivery

and longer duration of follow-up, for depressive relapse the four group studies with over 20 hours of therapy (5, 19, 20, 22) showed the most positive outcomes, with little heterogeneity (these are the top four studies in Fig. 3). By contrast, for manic relapse, the relationship between outcome and group delivery/hours of therapy is not clear, and an individual study delivering relatively few hours of therapy (focusing on early warning signs) was effective (17). Exploration of heterogeneity raised the possibility that, for manic relapse, the duration of follow-up may be more important than hours of therapy or group delivery. The benefit of early identification and treatment in preventing manic relapse is consistent with the efficacy of medication in treating elevated mood (1). By contrast, effective treatment of bipolar depression remains a major challenge (1), and even though Perry et al. (17) found that teaching patients to identify early warning signs of depression resulted in higher doses of antidepressants, this did not prevent depressive relapse. It is plausible that the reduction in depressive relapse lies in preventing episodes rather than their early treatment. Although the mechanisms by which psychoeducation might do this is not clear, candidates range from attitudinal impacts to changes in coping strategies and lifestyle (3). These are more likely to be better addressed in more extended therapies and by group delivery, as discussed above. As psychoeducation does not consistently improve depressive symptoms, it seems unlikely that a reduction in residual depressive symptoms, known to be associated with relapse (1), accounts for psychoeducation preventing depressive relapse.

Many aspects of psychoeducation are suited to web-based delivery. The single internet study showed no benefit on outcomes, not obviously explained by poor completion (18), but the limited amount of data produced means that conclusions about this mode of delivery cannot be drawn. Engagement with psychoeducation over the internet requires considerable motivation and can be a solitary process. An interesting question, if group delivery is an important factor in efficacy, is whether internet psychoeducation would benefit from being provided in a virtual group.

## Limitations

The relatively few studies, with differing methodologies and small participant numbers, raises caution about the results, which therefore need to be seen as raising questions rather than providing conclusive answers. There is most confidence in there

being a benefit for psychoeducation for any, and manic, relapse given that the true figure is likely to lie between the conservative and optimistic ITT analyses. Nevertheless, there is some concern that the findings are greatly dependent on the studies by Colom et al. (5, 20), which have unusually high control relapse rates. In order to address this, we carried out a sensitivity analysis of therapist-delivered studies excluding the studies by Colom et al., and still found a benefit for psychoeducation on all relapses, although the size of effect was reduced ( $NNT = 8$ ). Consistent with psychoeducation being clinically relevant, a recent study in a clinical service, not included in our analysis because it was not randomized, compared group psychoeducation with a matched waiting list arm and reported significantly fewer hospitalizations and days in hospital following psychoeducation over 12 months' follow-up (36). Given the small number of studies that have been carried out, there is insufficient evidence to draw firm conclusions about the comparative efficacy of psychoeducation against other specific psychotherapies for relapse prevention.

Patients with bipolar II disorder were in a minority in the studies that reported it, and therefore these results apply principally to bipolar I patients, with caution needed in extrapolating to those with bipolar II disorder. Nevertheless, a post-hoc analysis of the effect of psychoeducation over five years found that its efficacy in a subgroup of 20 patients with bipolar II disorder was comparable with that seen in the whole group (37).

We could potentially be criticized for our definition of psychoeducation and the limitation of the included studies to focus on the individual with bipolar disorder. In addition, our exclusion of studies with greater specific psychotherapy content could be argued to be unrealistic and narrow. However, the convergence of opinion and practice as to the content of psychoeducation in bipolar disorder provides credibility for it being a therapeutic approach in its own right. This is reflected in attempts to develop psychoeducation training and delivery programmes for mental health teams (38) and over the internet (18, 39) based on this model. Hence, we believe that it is justified to assess the efficacy of psychoeducation itself.

Finally, we cannot comment directly on the efficacy of psychoeducation delivered during active episodes of illness; however, the results from four RCTs identified in our search (39–42) (*Supplementary Table 1*) do not suggest that psychoeducation is effective in this situation.

#### Clinical implications

Group interventions based on the models of Colom and Vieta (32) or Bauer and McBride (33) have the best evidence for efficacy, making them currently the psychoeducation interventions of choice. Although the evidence is limited, psychoeducation appears to be effective for both bipolar I and bipolar II disorder. Individualized plans for seeking help based on early warning signs appears to be effective for preventing manic relapse, probably through enabling early treatment. Prevention of depressive relapse may require sufficient hours of therapy and a broader content, with the efficacy of brief psychoeducation interventions for this outcome unclear.

Involving family or carers may aid the early detection and treatment of elevated mood. The group study in which carers were involved was the only one to report a decrease in manic symptoms (13) and the comparative study of individually delivered psychoeducation against FFT (25) found that the severity and number of relapses were lower in the latter therapy involving family members.

The lack of evidence that internet-provided psychoeducation is effective suggests that it cannot be currently recommended as a stand-alone intervention.

#### Research implications

Although a longer duration of follow-up cannot be disentangled from other aspects, such as group delivery and hours of therapy in the studies reviewed, the most effective studies had study durations of over one year; this has some face validity as relapse may become more evident over a longer period.

A comparison of group and individually delivered psychoeducation needs to be tested in a clinical trial to inform practice, especially as individual psychoeducation is recommended by current UK national guidelines for patients with bipolar disorder (10); this should include a cost-effectiveness analysis as group delivery may be more economical even if efficacy is equal. In addition, there should be further research on shorter group interventions, to establish whether they can provide the degree of benefit seen with the longer interventions of Colom et al. (5, 20). Whether or not internet-delivered interventions are effective still requires establishing and it may be that novel designs are needed – for example, including interactive elements and encouraging learning from others with bipolar disorder.



Although an improvement in medication attitudes and adherence has been demonstrated with psychoeducation, the evidence that these persist is weak and needs to be tested in longer-term studies and related to clinical outcomes.

There is a need to identify which patients with bipolar disorder are likely to benefit from psychoeducation. Post-hoc analysis of the Colom et al. (5) study found that it was also effective in those with bipolar II disorder (37) and comorbid personality disorders (43). After five-year follow-up, the patients with the greatest all-round benefit were those with < 7 previous episodes, while no benefit was seen in those with > 14 episodes (44). Two key questions that need addressing are how to predict who will benefit from psychoeducation, and how to choose between psychoeducation and other types of psychotherapy for individuals with bipolar disorder. One approach may be to investigate how psychoeducation works – in particular, the effect of psychoeducation on attitudes to illness, cognition, lifestyle, and coping strategies, which has been little studied. Understanding the mechanisms involved could lead to better or more targeted interventions, help predict those at risk of relapse, and aid the personalization of treatment.

## Conclusions

Despite the limitations of, and heterogeneity in, the data, there is suggestive evidence that psychoeducation is effective in preventing relapse in bipolar disorder, with the strongest evidence for reducing overall and manic relapse. Greatest efficacy was found in studies with a group format, which also had a longer follow-up and more hours of therapy; these findings should inform clinical practice and guide future research. It is now important to investigate mediating mechanisms to be able to optimize efficacy and personalize treatment.

## Disclosures

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## Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Table S1.** Excluded studies.

**Table S2.** Description of psychoeducation interventions in controlled treatment trials in bipolar disorder.

**Figure S1.** Illustration of the interrelationship between psychological treatments for relapse prevention in bipolar disorder [based on Miklowitz et al. (2008) (11)].

**Figure S2.** Flowchart of search results.

**Figure S3.** Risk-of-bias graph.

**Figure S4.** Risk-of-bias summary.

**Figure S5.** Intervention dropouts comparing psychoeducation and placebo, active control, or treatment-as-usual.